

REVIEW! HEAT CONDUCTION

08/01/24

Heat Transfer vs Thermodynamics.

Fourier's law of heat transfer

Newton's law of cooling

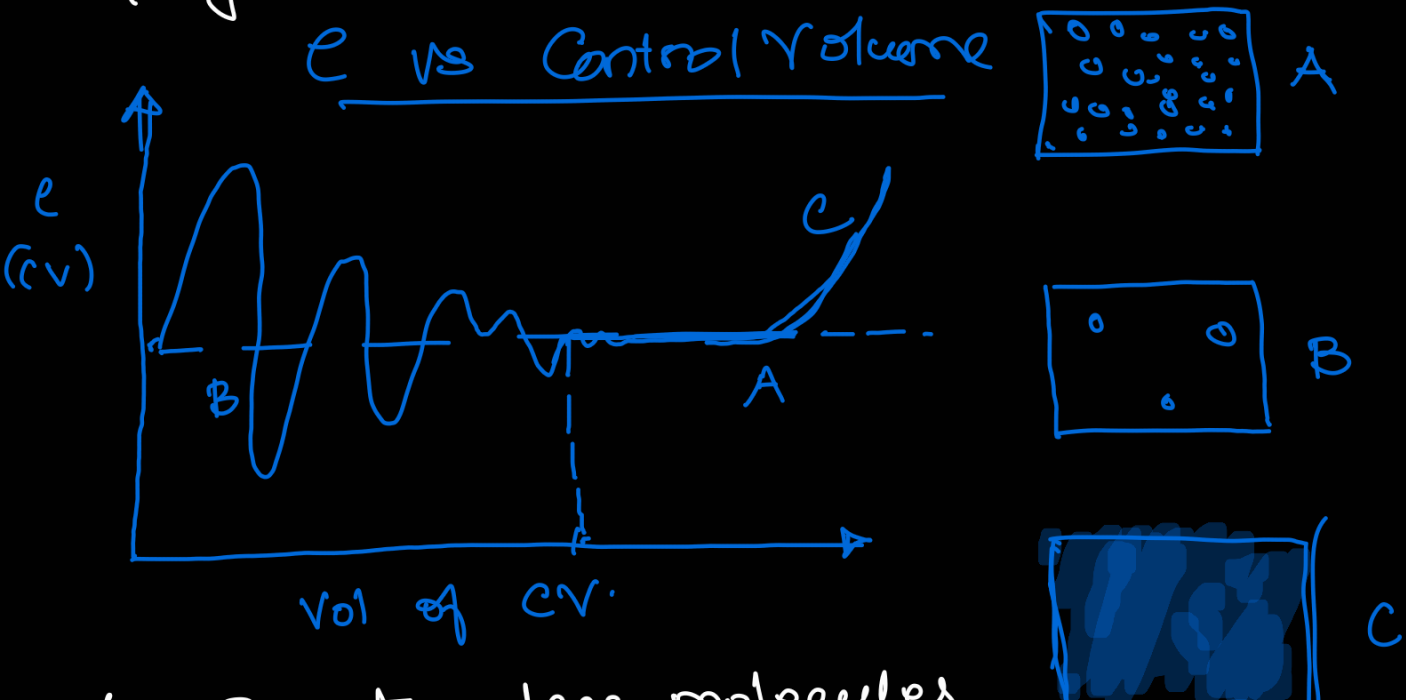
Stefan Boltzmann law of radiation.

CONTINUUM CONCEPT

Most fundamental approach: apply laws of motion and conduction to individual carrier molecules. A microscopic approach.

However for practical engineering applications microscopic approach is not required. It is better to assume the medium to be continuous i.e. continuum approach.

Continuum approach is only applicable when dimensions greater than mean free path length.



In B, too less molecules

hence density varies.

In A, enough molecules for density to remain constant.

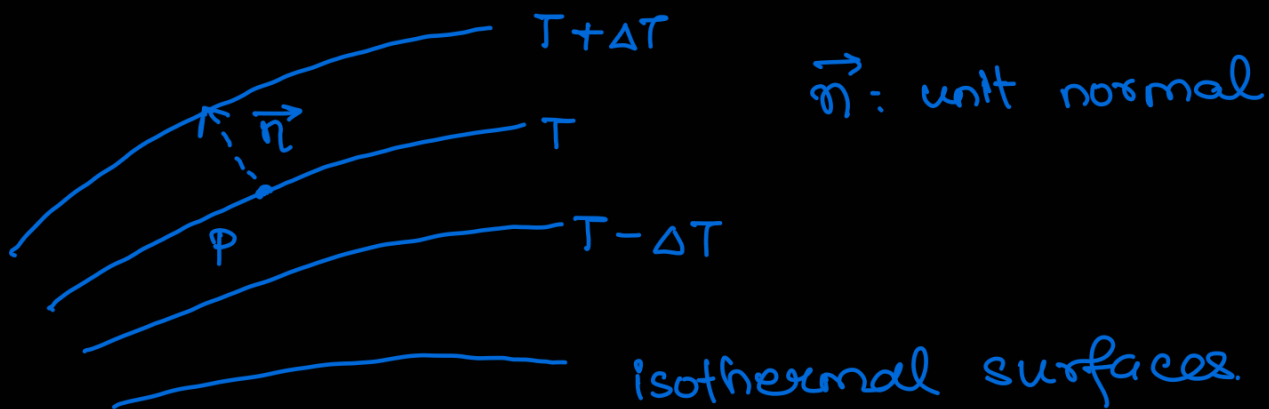
In C, too large that macroscopic density variation occurs.

Fourier's Law : $q'' = -k \frac{dT}{dx}$
(experimental law)

q'' : Heat flux

k : Thermal conductivity

Fourier's Law: Vector Form



$$q'' = -k \frac{\Delta T}{\Delta n}$$

$$= -k \left[\frac{\partial T}{\partial x} \cdot \frac{dx}{dn} + \frac{\partial T}{\partial y} \cdot \frac{dy}{dn} + \frac{\partial T}{\partial z} \cdot \frac{dz}{dn} \right]$$

$$\left\{ \frac{dx}{dn} = \cos \alpha \quad \frac{dy}{dn} = \cos \beta \quad + \quad \frac{dz}{dn} = \cos \gamma \right\}$$

α, β, γ are direction cosines

$$= -k \left[\frac{\partial T}{\partial x} \hat{i} + \frac{\partial T}{\partial y} \hat{j} + \frac{\partial T}{\partial z} \hat{k} \right] \cdot \vec{n}$$

$$q'_l = -k \nabla T \cdot \hat{n}$$

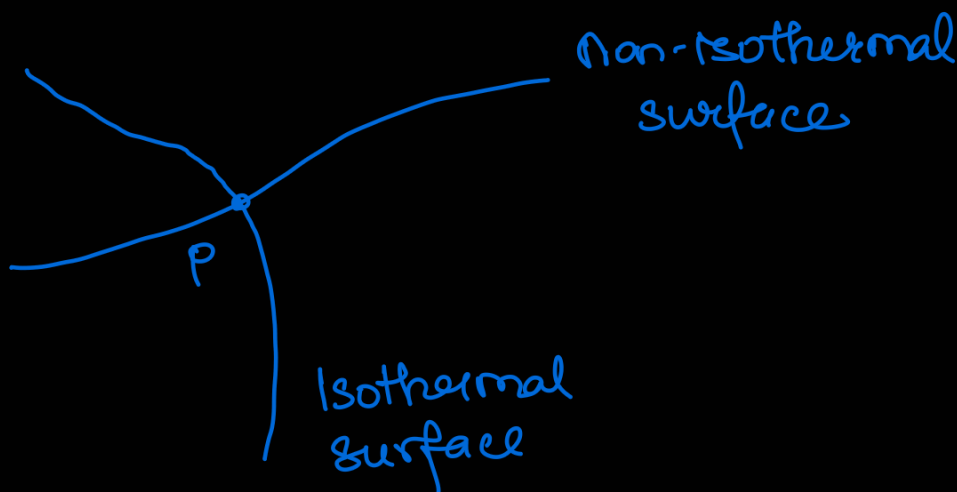
Cartesian Coordinates:

$$\nabla T = \hat{i} \frac{\partial T}{\partial x} + \hat{j} \frac{\partial T}{\partial y} + \hat{k} \frac{\partial T}{\partial z}$$

Cylindrical Co-ordinates:

Spherical Co-ordinates:

Extension of Fourier's law to Non-Isothermal Surface



$$q''_s = q'' \cdot \vec{s} = -k \nabla T \cdot \vec{s}$$

Since $\nabla T \cdot \vec{s} = \frac{\partial T}{\partial s} \Rightarrow q''_s = -k \frac{\partial T}{\partial s}$

Thermal Conductivity: Anisotropic Solid

Heat flux can be proportional to temp difference in other direction as well.

$$q''_1 = - \left\{ k_{11} \frac{\partial T}{\partial x} + k_{12} \frac{\partial T}{\partial y} + k_{13} \frac{\partial T}{\partial z} \right\}$$

$$q''_2 = - \left\{ k_{21} \frac{\partial T}{\partial x} + k_{22} \frac{\partial T}{\partial y} + k_{23} \frac{\partial T}{\partial z} \right\}$$

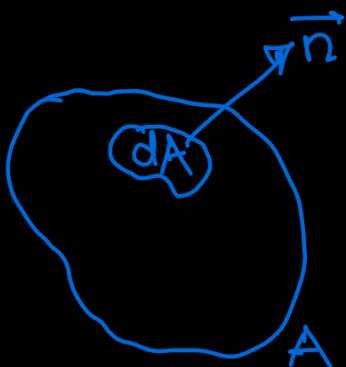
$$q''_3 = - \left\{ k_{31} \frac{\partial T}{\partial x} + k_{32} \frac{\partial T}{\partial y} + k_{33} \frac{\partial T}{\partial z} \right\}$$

$$K = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix} \quad \text{Thermal conductivity tensor}$$

$$\vec{q}'' = -\vec{K} \cdot \vec{\nabla} T$$

$$\vec{\nabla} T = \begin{bmatrix} \partial T / \partial x \\ \partial T / \partial y \\ \partial T / \partial z \end{bmatrix}; \quad \vec{q}'' = \begin{bmatrix} q''_1 \\ q''_2 \\ q''_3 \end{bmatrix}$$

TOTAL HEAT FLOW BY CONDUCTION



Rate of heat flux : $q'' \cdot \vec{n} dA$
through dA

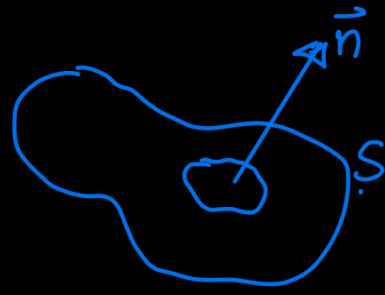
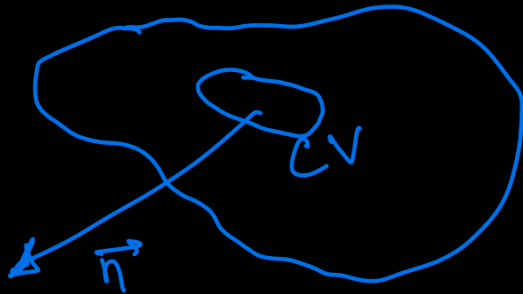
$$\text{Total heat flux: } \int_A q'' \cdot \vec{n} dA$$

q'' comes from Fourier's law.

DIVERGENCE THEOREM: $\oint_S \mathbf{F} \cdot d\mathbf{s} = \iiint_V \text{div}(\mathbf{F}) \cdot dV$
 surface integral volume integral.

HEAT CONDUCTION EQUATION

Stationary Opaque Solid



Control Volume

Energy Balance: $\frac{dE}{dt} = \dot{Q}$

$$\frac{dE}{dt} = \int_V \rho \frac{de}{dt} \cdot dV = \int_V \rho c_p \frac{\partial T}{\partial t} dV$$

$$\dot{Q} = - \int q'' \cdot \vec{n} \cdot dA + \int q''' dV$$

net amt of heat enter through surface heat generated within CV.

Converting surface integral to volume integral,

$$\int_V \rho c_p \frac{\partial T}{\partial t} \cdot dV = - \int_S q'' \cdot \vec{n} dA + \int_V q''' dV$$

$$\Rightarrow \int_V \left(\rho c_p \frac{\partial T}{\partial t} + \nabla q'' - q''' \right) dV = 0$$

$$\Rightarrow \rho c_p \frac{\partial T}{\partial t} + \nabla q'' - q''' = 0$$

substituting $q'' = -k \nabla T$ ($\nabla = \text{laplacian operator}$)

$$\nabla q'' = -k \nabla^2 T + \nabla k \cdot \nabla T$$

$$\rho c_p \frac{\partial T}{\partial t} = k \nabla^2 T + \nabla k \cdot \nabla T + q'''$$

$\nabla k \cdot \nabla T = 0$ for isotropic material (?)

Substituting the term, $\frac{\rho c_p}{k} = \frac{1}{\alpha}$

$$\boxed{\nabla^2 T + \frac{q}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}} \quad [\text{Fourier Biot Eqn}]$$

HEAT CONDUCTION : BOUNDARY CONDITIONS

Initial Condition: $T(r, t=0) = f(r)$

General BC: Conduction + Convection + Radiation.

$$-k \frac{\partial T}{\partial n} + h \Delta T + K_f (T^4 - T_0^4) (?)$$

For moving solids, add convective terms

$$\rho c \left[\frac{\partial T}{\partial t} + u \cdot \frac{\partial T}{\partial x} + v \cdot \frac{\partial T}{\partial y} + w \cdot \frac{\partial T}{\partial z} \right]$$

VARIABLE K : KIRCHOFF TRANSFORMATION

$$\nabla(k(T) \cdot \nabla T) + q(r, t) = \rho(T) c(T) \frac{\partial T}{\partial t}$$

Kirchoff Transformation, $\Theta(r, t) = \frac{1}{k_0} \int_{T_r}^T k(T) dT$

where T_r is the reference temp.
 $k_0 = k(T_r)$

$$\nabla^2 \Theta + \frac{q(r, t)}{k_0} = \frac{1}{\alpha(T)} \frac{\partial \Theta}{\partial t} \quad \left\{ \alpha(T) = \frac{k(T)}{\rho(T) c(T)} \right\}$$

$$\frac{\partial \theta}{\partial T} = \frac{k}{k_r} ; \quad \frac{\partial T}{\partial t} = \frac{\partial T}{\partial \theta} \cdot \frac{\partial \theta}{\partial t} = \frac{k_r}{k} \frac{\partial \theta}{\partial t}$$

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} = \frac{k_r}{k} \frac{\partial \theta}{\partial x}$$

Plugging into the eqn,

$$\left[\frac{\partial^2 \theta}{\partial x^2} + \frac{q'''}{k_r} = \frac{1}{\alpha} \frac{\partial \theta}{\partial t} \right]$$

$$\alpha = \frac{k(T)}{\rho C_p}$$

↳ one extra term is linear now.

Converting BCs in terms of θ as well,

